Health Insurance Data Analysis

Key Insights & Strategic Next Steps

KEY INSIGHTS – Risk Drivers

- Older age increases high-risk classification.
- Multiple chronic conditions significantly raise risk.
- Major procedures strongly linked to high-risk members.
- High Hb, LDL, and elevated blood pressure are key predictors.

Chronic Disease Burden

- Members with 3+ chronic conditions generate the highest claims.
- Most prevalent conditions: Arthritis, Hypertension, Diabetes, Cardiovascular disease, Mental health disorders.

Smoking Impact

- Smokers have higher medical costs, hospitalizations, and claim amounts.
- Risk increases further when combined with chronic diseases.

Age & Medical Costs

- Premiums and healthcare costs increase with age.
- Members aged 60+ have highest hospitalization and medication utilization.

Income & Coverage Gap

- Low-income members in high-deductible plans delay care.
- Higher emergency service usage among urban low-income members.

Geographic Gaps

- Rural members have fewer imaging and consultation visits (likely access-related).
- Urban members record higher lab tests and consultations.

Provider Quality Impact

- Higher provider quality scores correlate with fewer hospitalizations.
- Lower average claims indicate preventive care effectiveness.

Policy Churn Signal

- Members with 2+ policy changes in 2 years show higher claims activity.
- May indicate dissatisfaction or reactive enrollment.

NEXT STEPS – Predictive Risk Model

- Build ML model (XGBoost/LightGBM) on high-risk classification.
- Use SHAP values to identify key cost drivers.
- Integrate model into underwriting and renewal workflows.

Chronic Disease Program

- Assign care coordinators to top 5% high-risk members.
- Quarterly medication adherence outreach.
- Incentivize preventive checkups.

Low-Income Plan Redesign

- Introduce zero-deductible preventive care.
- Pilot sliding-scale copays.
- Launch rural community health worker program.

Smoking Cessation Program

- Cover Nicotine Replacement Therapy (NRT) at 100%.
- Partner with digital behavioral health platforms.
- Track cessation rates by region and plan type.

Improve Data Quality

- Alcohol frequency has ~30% missing data.
- Make it mandatory at enrollment.
- Apply MICE imputation for modeling.

Retain Policy Churns

- Identify members with frequent plan changes.
- Assign proactive retention team.
- Match members to better-suited plans before renewal.

Analytics Dashboard

- Track monthly claims trends.
- Monitor high-risk alerts.
- Detect procedure anomalies.
- Compare provider quality vs. cost benchmarks.